

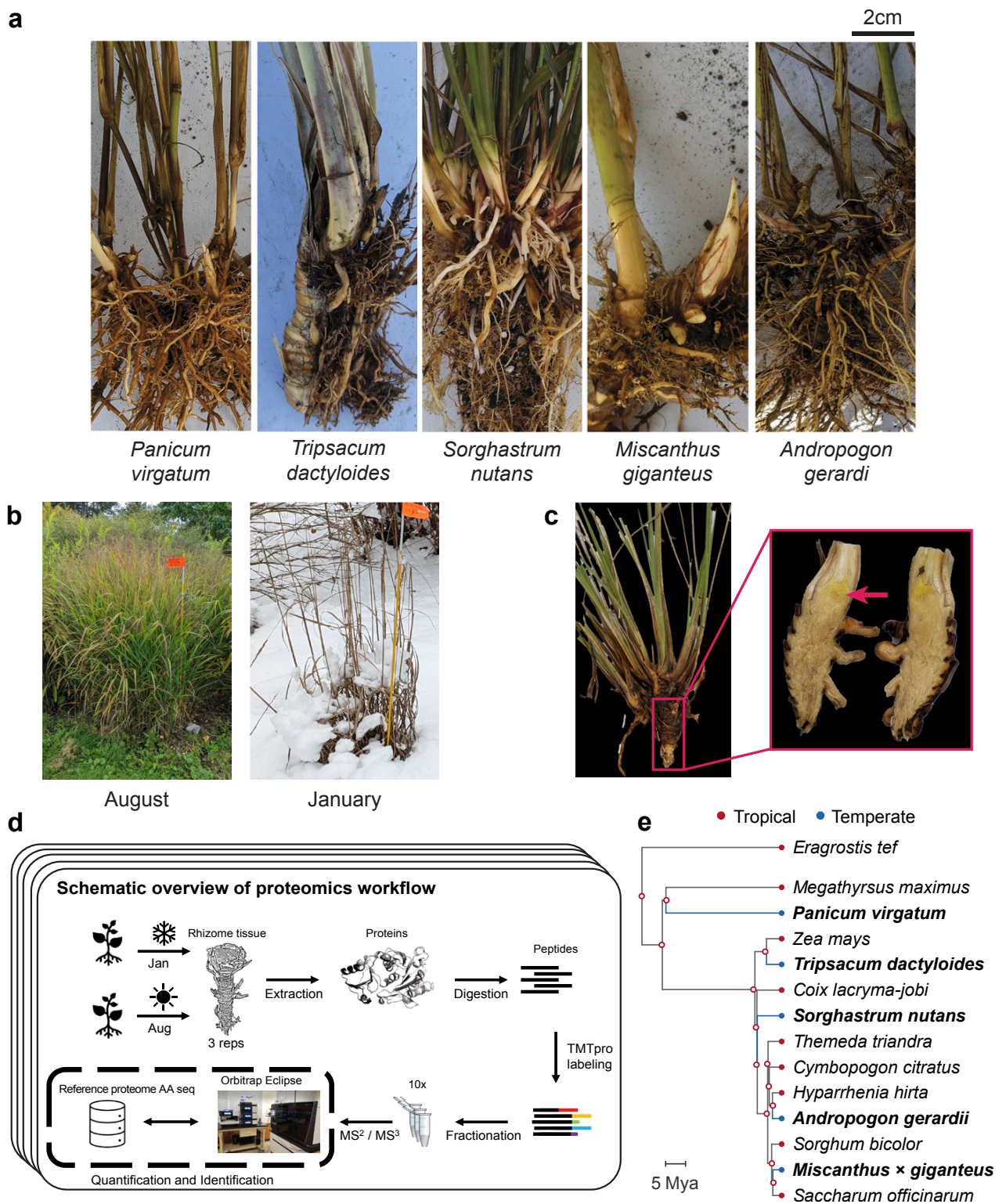


Fig. 1. Overview of sampling and proteomics workflow. (a) Rhizomes from the five species sampled in the study: *Tripsacum dactyloides* and *T. floridanum* hybrids, *Panicum virgatum*, *Andropogon gerardi*, *Miscanthus giganteus*, and *Sorghastrum nutans*. (b) Field conditions during the two sampling seasons, illustrating the active growth in August and dormancy in January. (c) Example of the rhizome tissue sampled for proteomics, shown here for *Tripsacum*. The red arrow highlights the specific region collected. (d) Schematic overview of the proteomics workflow. Rhizome samples were collected each season, followed by protein extraction, digestion, and TMTpro labeling. Peptides were fractionated and analyzed via RTS-SPS-MS³ using an Orbitrap Eclipse mass spectrometer. Each species' reference proteome was used for peptide and protein assignment, enabling quantification and identification. (e) Phylogenetic relationships among the five study species (bold) and tropical relatives, rooted on *Eragrostis tef* (Chloridoideae). Red, tropical; Blue, temperate and freezing tolerant.

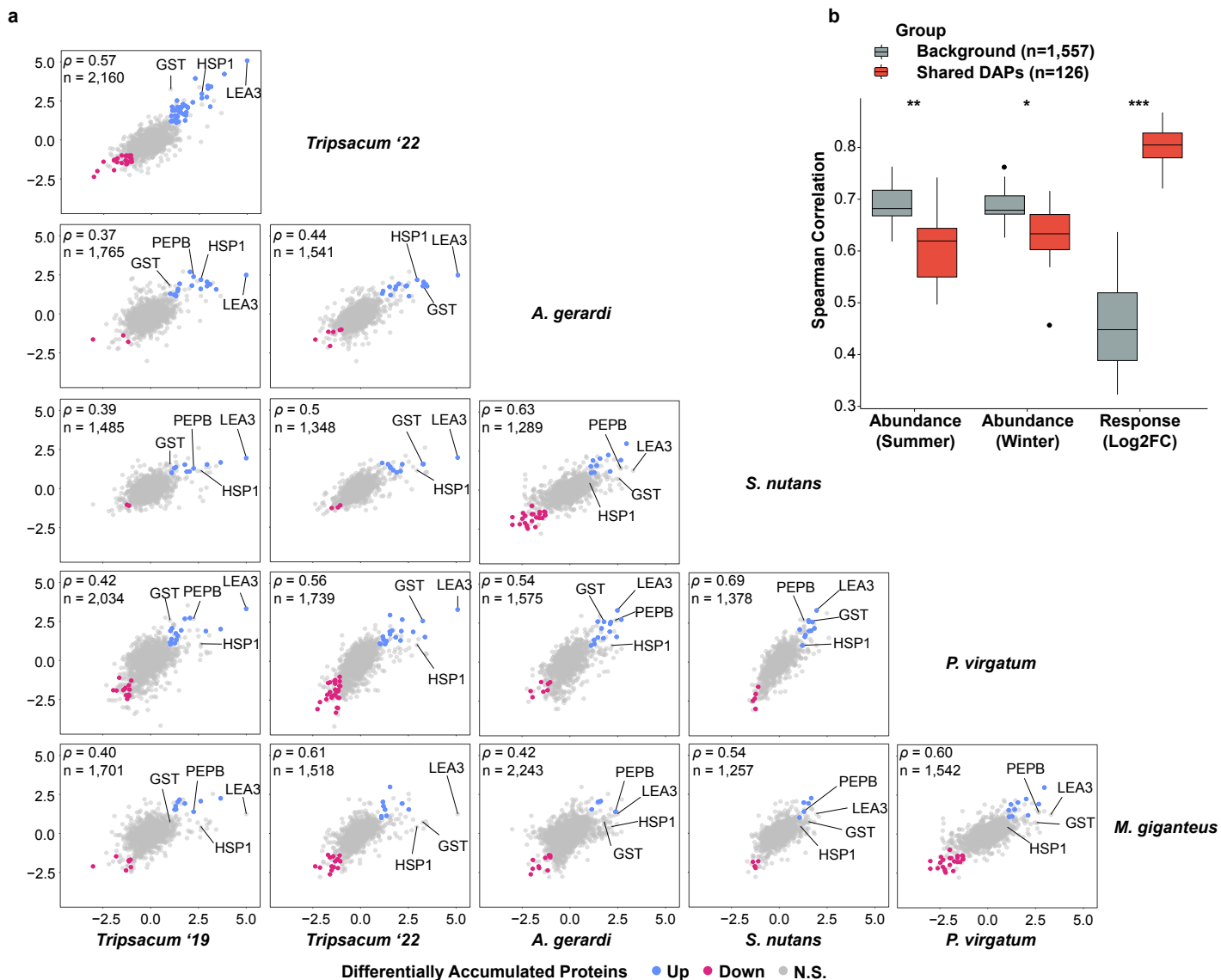


Fig. 2. Conservation of seasonal protein accumulation in rhizomes of *Tripsacum dactyloides* and *T. floridanum* hybrids, *Andropogon gerardi*, *Miscanthus* × *giganteus*, *Panicum virgatum*, and *Sorghastrum nutans*. (a) Pairwise comparisons of log₂ fold-change (log₂FC winter/summer) in differentially accumulated orthogroups across species. Scatterplots display the correlation of orthogroup-level log₂FC values between species, with shared DAPs elevated in winter shown in blue, reduced in red, and non-significant proteins in gray. Each panel represents a comparison between two species, with the number of shared orthogroups (*n*) and Spearman's correlation coefficient (ρ) displayed. (b) Pairwise Spearman correlations for orthogroups quantified in at least four of five species, comparing shared DAPs (red; differentially abundant in ≥ 2 species) versus background orthogroups (gray; differentially abundant in ≤ 1 species). Correlations are shown for summer abundance, winter abundance, and seasonal response magnitude (log₂FC winter/summer). Wilcoxon rank-sum test: **P* < 0.05, *P* < 0.01, ****P* < 0.001.**



Fig. 3. Overview of cold-related functional categories represented among the top 50 most winter-enriched rhizome proteins *Tripsacum dactyloides* and *T. floridanum* hybrids, *Andropogon gerardi*, *Miscanthus × giganteus*, *Panicum virgatum*, and *Sorghastrum nutans*. (a) Mosaic plot of cold-related functional categories across species. This plot visualizes the distribution of proteins among cold-related functional categories for each species. The relative abundance of each category within a species is shown, highlighting distinct functional profiles and species-specific variations in cold tolerance mechanisms. A Chi-square test for independence ($\chi^2[40] = 58.1$, $P = 0.032$) was performed to evaluate variation across species. (b) Visual representation of cold-related functional categories in rhizome proteins of five *Andropogoneae* species. This figure illustrates the top 50 most abundant proteins within the rhizomes of five *Andropogoneae* species. Proteins are classified into functional categories relevant to cold-related mechanisms, such as metabolism/osmoregulation, cold signal transduction, membrane stability, lipid metabolism, and others. Color-coding facilitates a comparative visualization of cold tolerance mechanisms across species. Antifreeze proteins are marked by an asterisk, as their classification is based solely on protein homology.

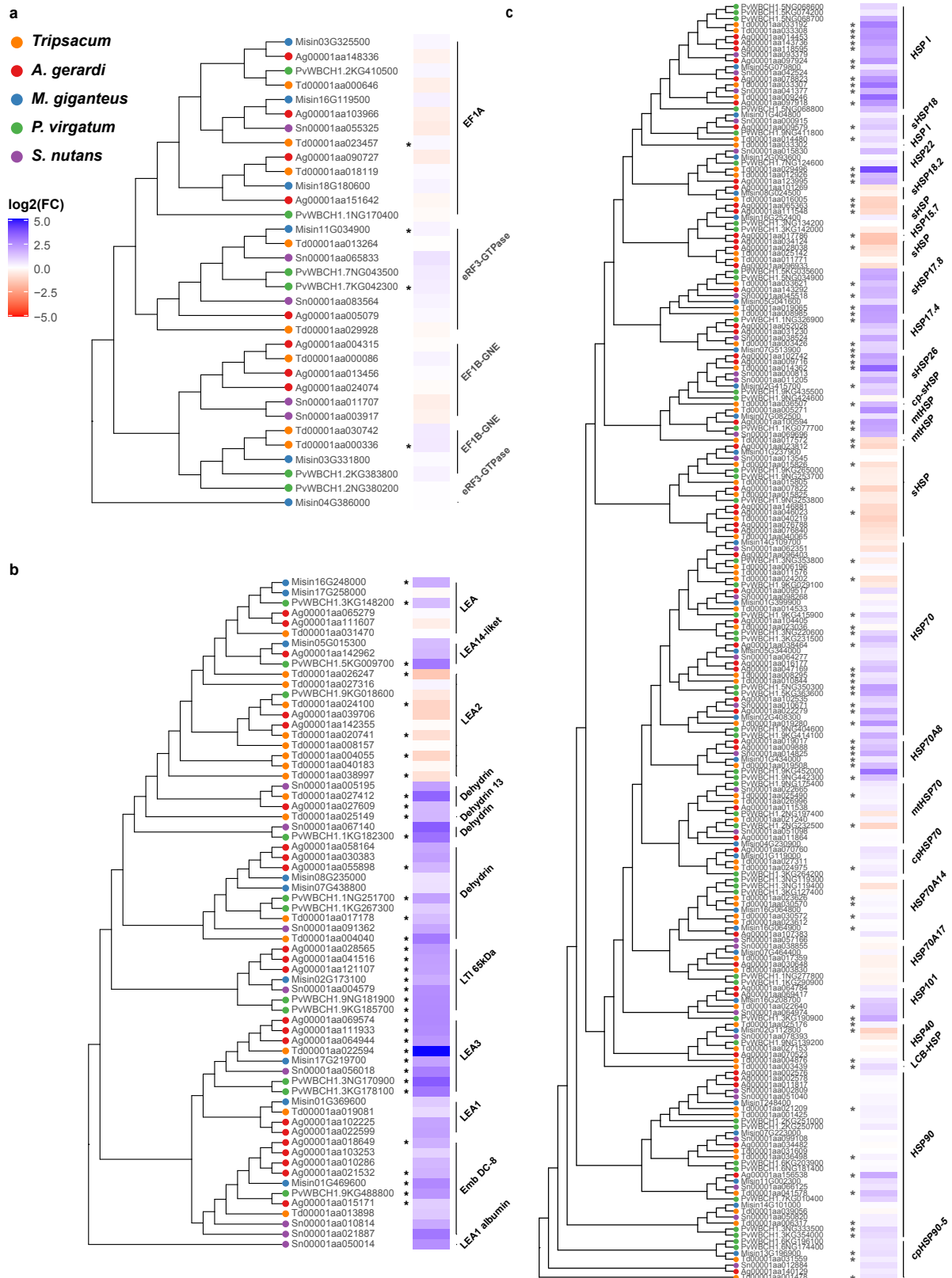


Fig. 4. Phylogenetic analysis and seasonal abundance dynamics of selected protein families across *Tripsacum dactyloides* and *T. floridanum* hybrids, *Panicum virgatum*, *Andropogon gerardi*, *Miscanthus* × *giganteus*, and *Sorghastrum nutans*. Protein identifiers represent species-specific sequences, and brackets denote functional group annotations, merging adjacent orthogroups with similar functions for clarity. Heat maps illustrate log₂ fold-change (log₂FC) in protein abundance between winter and summer conditions, where blue indicates higher abundance in winter, and red signifies higher summer abundance. Panels show distinct protein families: **(a)** translation elongation factor 1 proteins (EF1; control group) , **(b)** late embryogenesis abundant (LEA) proteins, and **(c)** heat shock proteins (HSPs). Asterisks (*) mark proteins exhibiting statistically significant differential accumulation ($P < 0.05$).

T. dactyloides MASHQDKASYQAGETKARTEEEKTGQAVGATKDTAQHAKDRASDAAGHA-----EATKQRAAETA-----EATKQRAAE

Z. mays MASHQDKASYQAGETKARTEEEKTGQAVGATKDTAQHAKDRAADAAGHAAGKGQDAKEATKQKASDTGSYLGGKTDEAKHKAGE

T. dactyloides TAEATKQRAAETAETATKQKAAETATAEATKQKAAE---YAKDTAVSGKDKSGGVIQQATEQVKSAAAGAKDAVMNTLGMGGDNKQ

Z. mays TTEATKQKQAGETTEAAKQKKAADAMEAAKQKAAEAGQYAKDTAVSGKDKSGGVIQQATEQVKSAAAGAKDAVMSTLGMGGDDKQ

T. dactyloides GDTDTTKDSSTITRDH

Z. mays GDANTNKDSSTITRDH

c Hydropath. / Kyte Doolittle

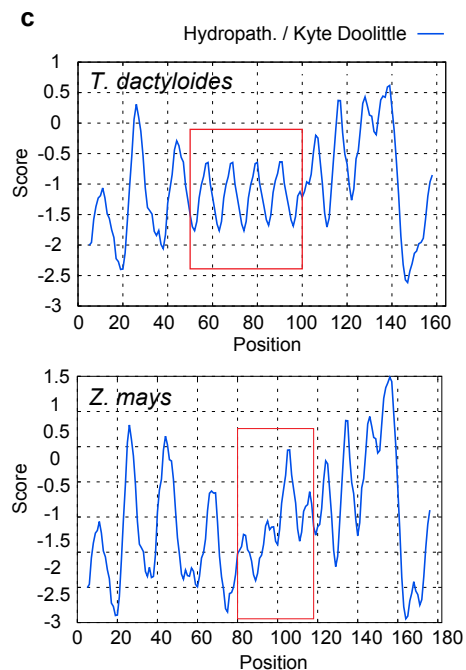


Fig. 5. Protein sequence alignment, structure and hydrophobicity of LEA3 orthologs from *Tripsacum dactyloides* and *Zea mays*. (a) The alignment shows the LEA3 proteins from *T. dactyloides* (Td00001aa022594_T002) and, *Z. mays* (Zm00001eb294480), with tandem 11-mer repeat units highlighted and labeled numerically. *T. dactyloides* contains five repeat units (1–5); *Z. mays* contains three (1–3). The dashed box indicates a repeat unit with variant residues relative to the consensus repeat sequence. (b) Predicted secondary structure of LEA3 proteins using AlphaFold3, showing the conserved alpha-helical domains across *T. dactyloides* (top) and *Z. mays* (bottom). Repeat regions corresponding to labeled sequence motifs are boxed. Red and blue indicate regions of high hydrophilicity and hydrophobicity, respectively. (c) Kyte-Doolittle hydropathy plots for LEA3 proteins. The X-axis represents residue position, while the Y-axis indicates hydropathicity scores (positive values for hydrophobic regions, negative for hydrophilic regions). Red boxes highlight the amphipathic region encompassing the 11-mer tandem repeat array, corresponding to key domains in (b).